Supply Chain Analytics - Demand Forecasting & Anomaly Detection

Project Overview

Developed an end-to-end Supply Chain Analytics solution using Oracle SQL, Python, and Power BI to analyze business performance, forecast future demand, and detect sales anomalies.

Tools Used

Oracle SQL, Excel, Python (Pandas, NumPy, Matplotlib, Statsmodels ARIMA), Power BI, DAX.

Project Steps

1. Collected and cleaned supply chain data.
2. Created SQL views for reporting.
3. Imported data into Power BI.
4. Built KPI measures using DAX.
5. Designed executive dashboards with cards, charts, and slicers.
6. Used Python ARIMA to forecast the next six months of demand.
7. Detected anomalies in sales data.
8. Imported forecast and anomaly outputs into Power BI.

Dashboard KPIs

Forecast Sales, Current Sales, Highest Forecast, Total Anomalies, Forecast Growth %, Demand Forecast Chart, Anomaly Detection Chart, Date Slicer.

Business Value

Supports inventory planning, supplier monitoring, sales trend analysis, anomaly detection, and data-driven decision making.

Skills Demonstrated

SQL, Excel, Python, Forecasting, ARIMA, Power BI, DAX, Data Visualization, Business Analytics.